

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 7: PROMPT REACTIVITY FEEDBACK PHENOMENA:

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

WHAT IS REACTIVITY FEEDBACK?

Reactivity feedback is the phenomenon that occurs when an originally applied reactivity changes the state of the system.

This change in the reactor causes a change in reactivity, $d\rho_{fb}$, which in turn feeds back to and modifies the original reactivity.

Reactivity feedback is called “inherent” if its occurrence is based on an unavoidable and thus totally reliable physical phenomenon.

*An example for “inherent” reactivity feedback is the **Doppler broadening** of resonances that is directly associated with the fuel temperature:*

- Doppler broadening of resonances automatically leads to an increase in neutron absorption and thus to a negative and inherent reactivity feedback.
- Note: The resonance capture dominates resonance fission in all power reactors.

“PROMPT” REACTIVITY FEEDBACK?

Reactivity feedback is called “prompt” if it directly follows the fuel temperature changes.

*If additional physical phenomena such as material motion or heat transfer are required to produce a feedback effect, **there will be a delay between the energy production in the fuel and the feedback.***

Note: *The delay in axial fuel expansion is small enough that this expansion can also be considered as a prompt feedback effect.*

REACTIVITY FEEDBACK IMPORTANCE FOR REACTOR SAFETY

If a reactor should become superprompt-critical, the neutron flux could (if no feedback) rise so rapidly that mechanical shutdown devices could not reduce the power before it reached a destructive level.

Fortunately, immediately counteracting shutdown phenomena (Doppler effect) that are inherent in the fuel rapidly reduces the super-prompt critical reactivity to subprompt-critical values and a rapid decrease in the reactor power results.

Consequently, the total energy release of such a transient can stay below damaging levels or cause only limited damage.

Since the counteracting reactivity feedback is inherent, the resulting shutdown effect is physically inevitable, and cannot fail to occur.

The impossibility of failure distinguishes inherent phenomena from man-made shutdown devices, where finite failure are probable.

Prompt Reactivity Feedback: Fuel Temperature Rise

PROMPT REACTIVITY FEEDBACK - **FUEL TEMPERATURE RISE**

The rise of the fuel temperature during a transient is due to the energy directly produced in fission reactions (~ 200 MeV per fission):

- *Kinetic energy of the fission products;*
- *Small additional contribution due to scattering and capture of fission neutrons;*
- *The emitted γ along with neutron capture (~ 5 to 7 MeV per capture and ~ 10 MeV per fission);*
- *Most of the original energy of the fission neutrons is lost in scattering (~ 5 to 6 MeV per fission);*
- *The radioactive decay of fission products.*

PROMPT REACTIVITY FEEDBACK - *FUEL TEMPERATURE RISE*

For an accurate treatment of feedback during transients, these contributions should be subdivided into:

- *Instantaneously released energy*
- *Delayed released energy*

as well as

- *a part deposited in the fuel*
- *a part deposited outside of the fuel material (cladding, coolant, moderator, and blanket or reflector)*

PROMPT REACTIVITY FEEDBACK - FUEL TEMPERATURE RISE

Instantaneously released energy – fission fragments

The major part of the energy released in the fission process ($\sim 83\%$) appears as a **kinetic energy of the two fission fragments**.

The fission fragments emerge as highly ionized atoms.

They lose their energy through ionization within a very small range ($\leq 10^{-2}$ mm in the fuel).

Ion pairs produced directly by the fission products also quickly dissipate their energy.

After a **negligibly short time ($\delta t \ll \Lambda$)** the energy of the fission fragments are converted into a temperature rise of a very small volume (fission track).

The range of fission products is so small compared to typical fuel rod diameters that **almost all of the energy of the fission fragments is deposited in the fuel**.

PROMPT REACTIVITY FEEDBACK - FUEL TEMPERATURE RISE

Instantaneously released energy – gamma-rays & neutrons

A smaller part of the total energy per fission ($\sim 11\%$) consists of practically “instantaneous” gamma rays ($\delta t \approx \Lambda$) and the energy directly deposited by neutrons in elastic scattering.

“Instantaneous” - the energy is deposited with an insignificant delay, i.e., during about one neutron generation time after its production.

As the mean free paths of neutrons are much larger than the fuel rod diameter, this part of the energy per fission is produced in all components of the core and even transported into the reflector or blanket.

The largest part of it is produced in the fuel (fission; inelastic scattering; and radiative capture).

The fuel has the highest reaction cross-section from gamma rays in the core - a large portion of this gamma-ray energy is also deposited in the fuel.

PROMPT REACTIVITY FEEDBACK - FUEL TEMPERATURE RISE

Delayed released energy

The third part of the fission energy, the decay heat appears with a delay that is significant in dynamics.

As a result of these delays and of the transport of some of the energy out of the fuel, ~10% less energy per fission is deposited in the fuel during a rapid transient than during a steady-state operation.

Conversion of the Number of Fissions into Heat

NUMBER OF FISSIONS TO HEAT

If C_{fsh} (“fission energy → stationary heat”) is the heat per fission in a stationary system, the stationary total power is written as:

$$P_0^{total} = C_{fsh} \int_V \int_E \Sigma_{f0} \phi_0 dE dV \qquad C_{fsh} \approx 0.35 \times 10^{-10} \text{ W} \cdot \text{s} / \text{fission}$$

These equations include the contribution of gammas from neutron capture and the delay energy release.

The power increase during a rapid transient is calculated with a different conversion factor, C_{fph} (prompt conversion of fission into heat)

If C_{fdh} describes the production of delayed heat (through decay of fission products), the following balance exists:

$$C_{fsh} = C_{fph} + C_{fdh}$$

Typical values obtained from experimental investigations of the fission process are:

$$\frac{C_{fph}}{C_{fsh}} \approx 0.94 \qquad \frac{C_{fdh}}{C_{fsh}} \approx 0.06$$

NUMBER OF FISSIONS TO HEAT

Fast transients - only the promptly produced heat to be considered.

Very slow transients - the delayed heat also must be considered.

Further complication - treatment of the heat directly deposited into the cladding and coolant through gamma rays or neutrons.

For now, only the prompt heat production in the fuel will be treated explicitly.

The conversion factor C_{fhf} denotes the corresponding constant, which converts the number of fission into promptly available heat in the fuel : $C_{fhf} \approx 0.9 C_{fsh}$

The remaining 10% of the heat production, delayed and/or deposited outside of the fuel rods, is not explicitly considered:

- *emphasis on dynamics with prompt reactivity feedback;*
- *accident analysis are concerned primarily with the core behavior.*

The 3% to 10% contribution leaving the core in the form of neutrons or gammas, or that is produced by fission in a FBR blanket normally has no effect on an accident progression.

*Core Power Models
and
Fuel Temperature Calculations*

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

Let $P(t)$ denotes the power of the core. We will consider transients starting from a stationary state.

The power transient $P(t)$ can then be split into the stationary power P_0 and the additional power $\delta P(t)$:

$$P(t) = P_0 + \delta P(t) \qquad P_0 = C_{fsh} \int_{core} \int_E \Sigma_{f0} \phi_0 dE dV \qquad \delta P(t) = C_{fhf} \int_{core} \int_E (\Sigma_f \phi - \Sigma_{f0} \phi_0) dE dV$$

$\text{UO}_2\text{-PuO}_2$ fuel has low heat conductivity $\rightarrow$ the heat transfer is fairly slow process $\rightarrow$ simplified heat transfer models.

During rapid transients, most of the energy deposited in the fuel raises the fuel temperature and is not transferred to the cladding or coolant $\rightarrow$ the heat transfer out of the fuel can be partially or fully neglected.

The neglect of fuel heat transfer, i.e., the adiabatic boundary condition of thermodynamics at the fuel cladding boundary, can be applied either to all of the heat or only to the incrementally produced heat.

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

Two formulas for the average fuel temperature rise, $\delta T(t)$, after the onset of a transients:

$$\delta T(t) = C_{QT} \int_0^t P(t') dt' \begin{cases} \text{completely} \\ \text{adiabatic fuel} \end{cases}$$

$$\delta T(t) = C_{QT} \int_0^t [P(t') - P_0] dt' \begin{cases} \text{adiabatic fuel for} \\ \text{power increment} \end{cases}$$

Adiabatic model for the incremental energy or “linear energy model” is:

- *based on the argument that the heat current out of the fuel rod is proportional to the negative gradient of the temperature distribution;*
- *identical to the adiabatic fuel model if the initial power is negligible compared to the power during the transient.*

The distribution of the additionally produced heat is essentially flat over the rod:

During a rapid transient, the temperature distribution is merely lifted up without any significant change in the shape of the distribution.

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

The adiabatic model can be improved in a first approximation by treating the heat release out of the fuel with a single time constant, λ_H :

$$\delta Q_{\text{remaining in fuel}} = \int_0^t [P(t') \exp[-\lambda_H (t - t')]] dt'$$

$$\delta Q_{\text{total}} = \int_0^t \delta P(t') dt'$$

$$\delta Q_{\text{released from fuel}} = \int_0^t \delta P(t') \{1 - \exp[-\lambda_H (t - t')]\} dt'$$

The heat release constant λ_H for a cylindrical fuel rod of radius R with uniform heat production rate and temperature-independent heat conductivity is proportional to $1/R^2$:

$$\lambda_H \propto \frac{1}{R^2}$$

Example: for a 6-mm (~0.25-in)-diameter oxide fuel rod the heat release constant becomes $\lambda_H \approx \frac{1}{2s}$

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

The energy-temperature conversion constant for the fuel is:

$$C_{QT} \left[\frac{K}{MW \cdot s} \right] = \frac{1}{c_h} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \frac{1}{V_f [mm^3]} \cdot \frac{W}{MW}$$

The fuel heat capacity per unit volume, c_h , is given by the product of the specific heat, c_p , and density, ρ_d :

$$c_h \left[\frac{W \cdot s}{K \cdot mm^3} \right] = c_p \left[\frac{W \cdot s}{K \cdot g} \right] \rho_d \left[\frac{g}{mm^3} \right]$$

If δQ_{cf} is the energy deposited in the core fuel, the corresponding temperature increase is:

$$\delta T(t) = C_{QT} \delta Q_{cf}(t)$$

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

If q_{cf} is the energy density deposited in the fuel and C_{qT} is the corresponding conversion factor, the temperature increase is:

$$\delta T(t) = C_{qT} V_f \frac{\delta Q_{cf}(t)}{V_f}$$

$$\delta T(t) = \frac{1}{c_h} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \delta q_{cf} \left[\frac{W \cdot s}{mm^3} \right]$$

$$\delta T(t) = C_{qT} \delta q_{cf}(t)$$

For the calculation of $\delta T(t)$ it is convenient to introduce a separation of power into the nominal power, P_n (usually called “full power”), and the flux amplitude function $p(t)$:

$$P(t) \approx P_n p(t)$$

$$\delta T(t) = C_{qT} \delta Q_{cf}(t)$$

$$\delta Q_{\text{total}} = \int_0^t \delta P(t') dt'$$

$$\delta T(t) = \frac{1}{c_h} \frac{P_n [W]}{V_f [mm^3]} \int_0^t \delta p(t') dt'$$

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

The integral over the amplitude difference, $\delta p(t) = p(t) - p_0$, can be introduced as:

$$I(t) = \int_0^t [p(t') - p_0] dt'$$

The integral $I(t)$ describes the additional energy in dimensionless units, which are called here “relative full-power seconds”

For example, $I(t) = 1.5$ gives an energy release of 1.5 full-powers second (fp-s)

The conversion factor is denoted by C_{IT} , in analogy with other notations: $\delta T(t) = C_{IT} I(t)$

The important conversion factor C_{IT} has dimension of K/s and it is the rate of temperature rise per second of (uncooled) nominal power application:

$$C_{IT} [\text{K/fp} \cdot \text{s}] = \left(\frac{\partial T}{\partial t} \right)_{P_n}$$

Note that “fp-s” has a dimension of time and not a dimension of energy as the product $P_n \cdot \text{second}$ or fp·s:

$$C_{IT} [\text{K/s}] = C_{IT} [\text{K/fp} \cdot \text{s}]$$

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

For example, for oxide fuel with an operational power density of 0.13 or 1.3 MW/liter of fuel (0.13 or 1.8 W/mm³), typical of BWRs or FBRs, respectively

$$C_{IT} = \frac{1}{0.0032} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \cdot \begin{cases} 0.13 \text{ W/mm}^3 \\ 1.3 \text{ W/mm}^3 \end{cases}$$

$$C_{IT} \approx \begin{cases} 40 \text{ K/fp} - \text{s} \\ 400 \text{ K/fp} - \text{s} \end{cases}$$

For PWRs, C_{IT} is about twice as large as for BWRs

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

These relations hold for the average fuel temperature in the core, which was denoted by T for simplicity.

The fuel temperature within a fuel rod varies strongly around this average.

The fuel rod averaged temperature also depends on space.

The “average” core fuel temperature, T , is generally used as basis for the calculation of prompt reactivity feedback.

For the fairly flat power distribution across the fuel rods, the temperature distribution within the rod is approximately given by a parabola in the radial coordinate r :

$$T(r) = T_0 - \frac{r^2}{R^2}(T_0 - T_R)$$

where R designates the fuel radius, and T_0 and T_R the temperatures at $r = 0$ and $r = R$, respectively.

The rod average value, $\bar{T}_{rod}$, can be obtained from the above Equation as $\bar{T}_{rod} = (T_0 + T_R)/2$

CORE POWER MODELS AND FUEL TEMPERATURE CALCULATIONS

The Equation for $T(r)$ is derived under the assumption of temperature-independent thermodynamic properties and a space-independent heat production rate within a fuel rod.

The temperature variation within the fuel rod described by this equation is quite significant due to the high power density and due to the low heat conductivity of the oxide fuel.

In fast reactors, $(T_0 - T_R)$ may be larger than 1000 K across an oxide fuel rod, but only about 100 K for metal fuel.

The Doppler reactivity feedback effect is generally calculated for media with a uniform temperature distribution and not for the strongly space-dependent temperature profile within a fuel rod.

An effective temperature therefore must be found that yields the correct Doppler reactivity when it is inserted into the formula based on a uniform temperature distribution.

Next Class *Transient at Small Times*